Static Structural Analysis and Safety Factor Evaluation of a Twin-Wheel Nose Landing Gear Assembly using Finite Element Analysis

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***Abstract*—Landing gear assemblies are among the most critically loaded structures on an aircraft, required to absorb large impact energies while remaining within tightly constrained mass budgets. This paper presents an independent design and finite element analysis (FEA) study of a twin-wheel nose landing gear (NLG) assembly, modeled as a multi-body CAD assembly and evaluated under representative static landing loads using Ansys Mechanical. The assembly, consisting of a telescopic oleo strut, a scissor (torque-link) mechanism, an axle fork, and dual wheels, was constrained at its upper trunnion mount and subjected to three orthogonal component loads representing vertical touchdown reaction, side (drag/spin-up) loading, and strut-axial loading. A tetrahedral finite element mesh was generated and the resulting equivalent stress, elastic strain, total deformation, and safety factor fields were computed. The analysis identifies the fillet region of the upper torque link as the critical location, returning a minimum static safety factor of 0.667, indicating that the current geometry does not meet a factor-of-safety of unity under the assumed loading and is a candidate for local geometric reinforcement. The study demonstrates a complete, self-directed CAD-to-FEA workflow undertaken as an undergraduate passion project and outlines a roadmap of refinements – mesh convergence, material substantiation, and dynamic drop-test simulation – required to mature the design toward airworthiness-representative standards.**

***Index Terms*—landing gear, finite element analysis, static structural analysis, safety factor, torque link, Ansys Mechanical, CAD**

I. Introduction

The landing gear is the single structural subsystem of an aircraft that must reliably transfer the entire kinetic energy of touchdown into the airframe without exceeding allowable stress limits, while simultaneously enabling ground steering, taxiing, and shock absorption [1]. Nose landing gear (NLG) assemblies, in particular, combine a telescopic shock-absorbing strut with a scissor (anti-rotation torque-link) mechanism that prevents relative rotation between the sliding piston and the fixed outer cylinder while still permitting axial stroke.

This work was undertaken independently, outside coursework, to build practical competence in the complete product-design pipeline: solid modeling of a multi-body mechanical assembly, definition of realistic boundary conditions, meshing, static structural solution, and post-processing of stress and safety-factor fields. The motivation stems from an interest in aerospace structures and production engineering, and the exercise is treated here as a first-principles case study rather than a certified design.

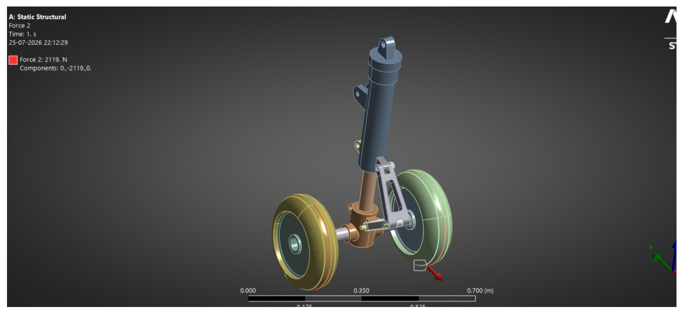

Fig. 1: Isometric render of the nose landing gear assembly as loaded in Ansys Mechanical, showing the outer strut, scissor links, piston, shaft, and twin wheel-hub/wheel units (highlighted).

The remainder of this paper is organized as follows. Section II describes the seven-part CAD assembly, the anatomy of each individual component, and how they mate kinematically. Section III details the finite element methodology, including material assumptions, meshing strategy, and the applied boundary conditions. Section IV presents the simulation results. Section V discusses the findings, their limitations, and future work, and Section VI concludes the paper.

II. Component Anatomy and Assembly

The landing gear was modeled as a seven-part SolidWorks assembly (`Assembly.SLDASM`), built from seven individually authored part files – `Strut.SLDPRT`, `Piston.SLDPRT`, `Shaft.SLDPRT`, `Upper_Link.SLDPRT`, `Lower_Link.SLDPRT`, `Wheel_Hub.SLDPRT`, and `Wheel.SLDPRT` – and imported into Ansys Mechanical (2026 R1, Student release) for analysis. Fig. 1 shows the assembled isometric render, and Fig. 2 shows a labeled schematic identifying each of the seven parts and the joints connecting them.

A. Individual Part Anatomy

- **Strut** – the outer telescopic cylinder and the primary load path to the airframe. Its upper end forms the trunnion lug that is bolted to the aircraft structure (and is the fixed-support face used in Section IV); its lower bore houses the sliding Piston and provides the upper pivot lug for the Upper_Link.
- **Piston** – the inner telescopic rod that slides axially inside the Strut bore, forming the oleo-pneumatic shock-absorbing degree of freedom. Its lower end widens into an integral fork/clevis feature that cradles the Shaft, and a mid-length lug provides the lower pivot point for the Lower_Link.
- **Upper_Link** – the upper arm of the scissor (anti-rotation torque-link) mechanism. It is pinned to the Strut at one end and to the Lower_Link at the other, forming the knuckle joint of the scissor pair. This part contains *Fillet3*, the feature identified in Section V as the location of minimum safety factor.
- **Lower_Link** – the lower arm of the scissor mechanism, pinned to the Piston at one end and to the Upper_Link at the knuckle. Together, Upper_Link and Lower_Link transmit torque between Piston and Strut while allowing the telescopic stroke to change their included angle.
- **Shaft** – the axle that passes through the Piston fork and through the bore of each Wheel_Hub, reacting the vertical and lateral wheel loads (Section IV-C) into the Piston.
- **Wheel_Hub** – the machined hub that mounts onto the Shaft and provides the bolted/bearing interface to the Wheel, transferring radial and axial tire loads onto the Shaft.
- **Wheel** – the tire-and-rim component, instantiated twice (as a mirrored pair) to form the dual-wheel arrangement common to nose landing gear, providing redundancy and distributing the ground-contact patch load.

B. Assembly and Kinematics

The seven parts mate through three functional joint groups, illustrated in Fig. 2. First, the **Strut–Piston** pair forms a cylindrical (sliding) joint along the vertical axis, representing the oleo-pneumatic stroke. Second, the **scissor linkage** – Strut–Upper_Link–Lower_Link–Piston – forms a four-bar pinned mechanism in which each of the three inter-part joints (Strut/Upper_Link, Upper_Link/Lower_Link, Lower_Link/Piston) is a revolute pair; this mechanism kinematically constrains the Piston against rotating relative to the Strut while still permitting it to translate, a classical arrangement in oleo-pneumatic nose gear design [1], [2]. Third, the **Shaft–Wheel_Hub–Wheel** group forms the rotating wheel axis: the Shaft is captured in the Piston fork, the two Wheel_Hub parts are mounted concentrically on the Shaft, and the two Wheel parts are mounted onto their respective hubs, giving the twin-wheel unit that contacts the ground. In the Ansys Mechanical model these mating relationships were carried over from the native SolidWorks assembly constraints and resolved as bonded/contact pairs for meshing and solution.

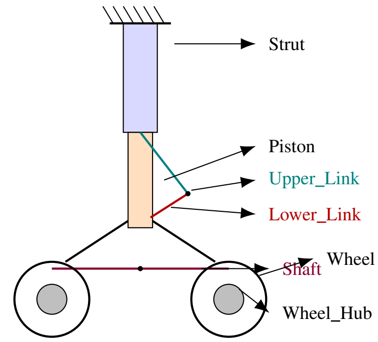

Fig. 2: Labeled kinematic schematic of the seven-part assembly: the Strut–Piston cylindrical (sliding) joint, the Strut–Upper_Link–Lower_Link–Piston scissor mechanism, and the Shaft–Wheel_Hub–Wheel rotating axle group.

C. Individual Part Models

Fig. 3 shows each of the seven native SolidWorks part models rendered individually, prior to assembly. The Strut and Piston are primarily revolved/extruded cylindrical bodies; the Upper_Link and Lower_Link are thin-walled extruded forks with lightening cut-outs and filleted lugs at each pin location; the Shaft is a stepped, filleted turned part; and the Wheel_Hub is a machined, multi-axis boss-and-bore part that also carries the Nickel material assignment used for this component. The Wheel part combines an imported tire profile with a rim body to form a two-body part.

D. Exploded View and Final Assembly

Fig. 4 shows an exploded view of the assembly (generated in eDrawings), which visually separates the seven parts along their assembly axes and confirms the mating scheme described in Section II-B: the Strut and Piston separate along the telescopic axis, the Upper_Link/Lower_Link pair separates away from their pin locations, and the Shaft, Wheel_Hub, and Wheel units separate along the axle axis. Fig. 5 shows the fully mated assembly, including the flange-bolt fasteners used at the Strut–Upper_Link, Piston–Lower_Link, and Shaft–Wheel_Hub pin joints, as modeled in the SolidWorks assembly feature tree.

III. Finite Element Methodology

A. Material Model

For this baseline study, six of the seven parts (Strut, Piston, Shaft, Upper_Link, Lower_Link, Wheel) were left at the default linear-elastic isotropic structural steel material available in the Ansys Engineering Data library, while the Wheel_Hub was separately assigned a Nickel material in its native SolidWorks part file. This mixed, largely default assignment is a simplifying choice made for a first-pass static check; production landing gear members are typically manufactured from high-strength alloy steels (e.g., 300M) or aerospace aluminum/titanium alloys, and substituting a consistent, alloy-specific material set is identified in Section V as a required next step.

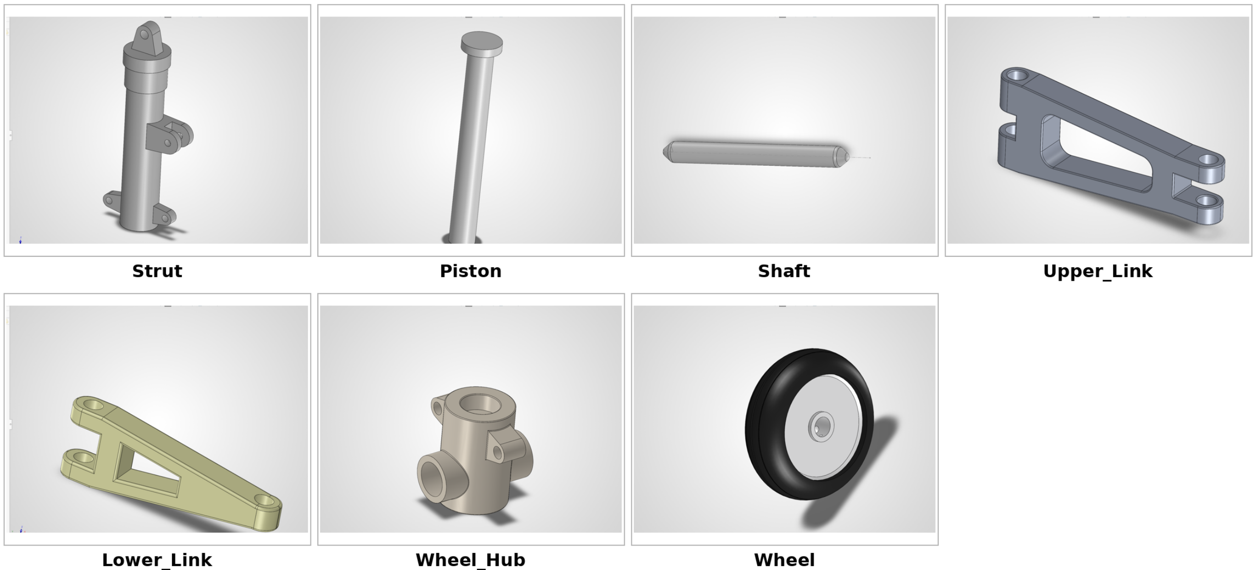

Fig. 3: Individual SolidWorks part models comprising the assembly: Strut, Piston, Shaft, Upper_Link, Lower_Link, Wheel_Hub, and Wheel.

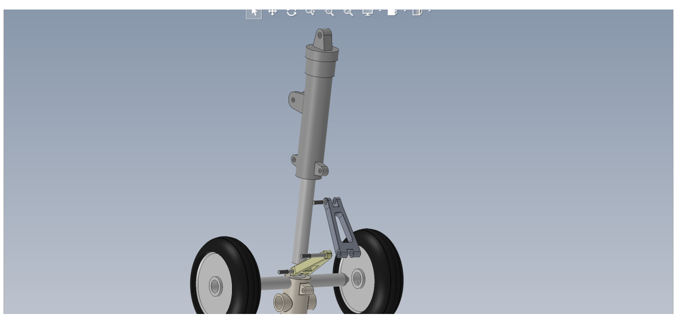

Fig. 4: Exploded view of the assembly (eDrawings), showing the Strut–Piston telescopic separation, the Upper_Link/Lower_Link scissor pair, and the Shaft–Wheel_Hub–Wheel axle group.

B. Meshing

A tetrahedral (patch-conforming) mesh was generated over the full assembly using the Ansys Mechanical default sizing algorithm with distributed solve across four cores, as visible in the wireframe overlay of Figs. 11–10. No local mesh refinement was applied at the Upper_Link/Lower_Link fillets or the Shaft bearing surfaces in this iteration, which is a known source of localized stress over-prediction discussed in Section V.

C. Boundary Conditions and Loading

The assembly was constrained with a *fixed support* applied to the single trunnion face at the top of the Strut, representing rigid attachment to the airframe (Fig. 7). This single fixed reference is consistent with treating the airframe attachment as infinitely rigid relative to the gear itself – a standard simplification for an isolated-component static check.

Fig. 5: Final mated assembly in SolidWorks, showing the seven parts joined with flange-bolt fasteners at each pin location.

TABLE I: Applied Static Load Cases

Load	Magnitude (N)	Direction	Scope
Force (axle vertical)	7062	$+Z$	2 edges (axle)
Force 2 (side / drag)	2119	$-Y$	2 edges (axle)
Force 3 (strut axial)	15696	$+Z$	1 face (piston)

Three surface-effect force loads were applied to represent a static ground/landing reaction case, summarized in Table I and illustrated schematically in Fig. 6:

All loads were ramped linearly from zero to their full magnitude over a single time step, consistent with a quasi-

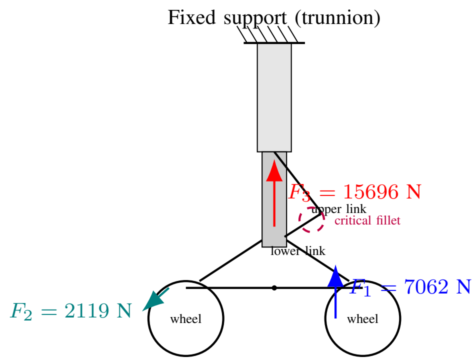

Fig. 6: Simplified free-body schematic of the nose landing gear showing the fixed trunnion constraint, the three applied static load components (F_1: axle-vertical, F_2: axle-side, F_3: strut-axial), and the location of the minimum-safety-factor fillet on the upper torque link.

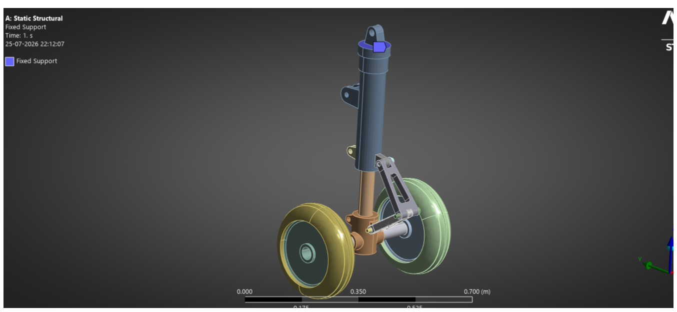

Fig. 7: Fixed support applied to the upper trunnion face of the outer strut cylinder.

static solution philosophy. *Force* and *Force 2* act on the wheel axle edges, representing the vertical touchdown reaction and a lateral drag/side load respectively, while *Force 3* acts directly on the piston face, representing the axial reaction transmitted up the shock strut. Together, the three loads approximate a combined vertical-landing-plus-side-load case of the type used in preliminary landing-gear strength checks [1], [4].

IV. Results

A. Boundary Condition and Load Verification

Figs. 7–10 show the applied fixed support and the three force loads as rendered directly in Ansys Mechanical, confirming correct scoping (face/edge selection), direction, and ramped magnitude for each load prior to solution.

B. Safety Factor Distribution

Fig. 11 shows the resulting static safety factor contour over the full assembly. The safety factor ranges from a maximum of 15 (capped) down to a minimum of **0.667**, occurring on the fillet of the upper torque link (reported by the solver as *Upper Link|Fillet3*). A safety factor below unity in a linear-elastic static check indicates that, under the assumed load case and material, the computed stress at that location exceeds the yield strength of the assigned material, i.e. the component would be predicted to yield locally before the full load case is reached.

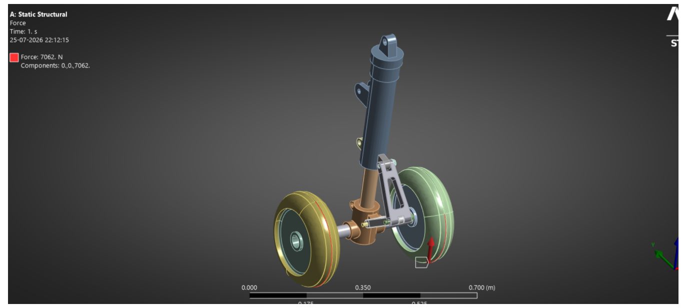

Fig. 8: Force load of 7062 N applied along $+Z$ on the wheel axle edges, representing the vertical landing reaction.

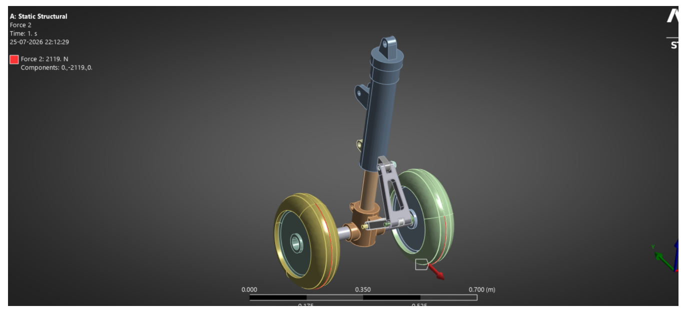

Fig. 9: Force 2 load of 2119 N applied along $-Y$ on the wheel axle edges, representing a lateral drag/side load.

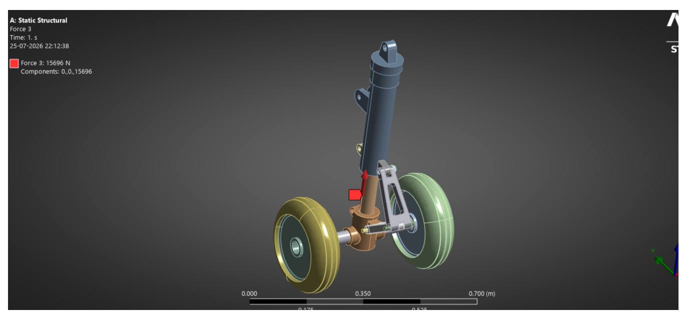

Fig. 10: Force 3 load of 15696 N applied along $+Z$ on the piston face, representing the axial strut reaction.

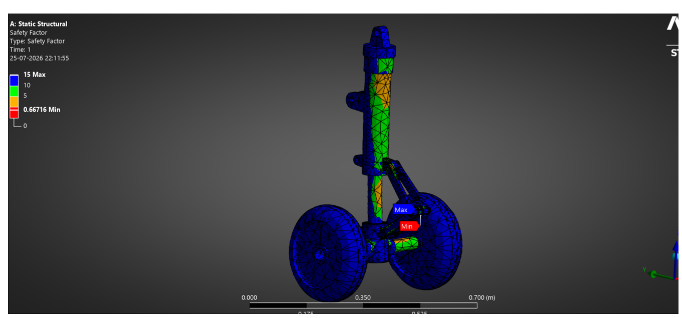

Fig. 11: Static safety-factor contour for the assembly. Minimum SF = 0.667 occurs at the upper torque-link fillet; maximum plotted value is capped at 15.

TABLE II: Summary of Static Structural Results

Quantity	Value
Minimum safety factor	0.667
Maximum safety factor (capped)	15.0
Average safety factor	13.63
Location of minimum SF	Upper Link, Fillet3
Total applied vertical + axial load	22758 N
Total applied lateral load	2119 N

C. Summary Table

Table II summarizes the key scalar results extracted from the solution.

V. Discussion

The minimum safety factor of 0.667 at the upper torque-link fillet is the single most important outcome of this study. Two competing interpretations are worth noting. First, it may reflect a genuine local structural deficiency – torque-link fillets are classical stress-concentration features, and if the modeled fillet radius is small relative to the link cross-section, elastic stress concentration factors of 2–3$\times$ nominal stress are physically plausible [1]. Second, because no local mesh refinement was applied at this feature (Section III-B), the reported peak stress is also likely inflated by mesh-density-dependent stress singularity at the sharp re-entrant corner; unconverged fillet stresses are one of the most common sources of overly conservative safety factors in first-pass FEA studies. Distinguishing between these two explanations requires a mesh convergence study, in which the fillet radius region is progressively refined until the reported peak stress stabilizes to within a few percent between successive refinements.

A second limitation is the material assignment. The default structural steel used here is a reasonable placeholder for a first static check but does not represent the certified aerospace alloys (e.g. 300M steel, 15-5PH stainless, or 7075-T6/Ti-6Al-4V for lighter members) used in production landing gear, whose yield strengths can differ by a factor of two or more from generic structural steel. Re-running the analysis with an alloy-specific material curve is a direct and low-effort next step.

Third, the load case itself is a static approximation of what is, in reality, a highly dynamic event. Real landing gear certification load cases are derived from drop-test energy absorption requirements and dynamic load factors specified in airworthiness standards such as 14 CFR §25.473 [4], rather than from three static point loads. A natural extension of this work is to model the oleo-pneumatic shock absorber's force-stroke behavior explicitly and perform a transient/drop-test simulation to obtain a more representative peak dynamic load spectrum, which could then be fed back into the static strength check as a validated equivalent static load.

Future work arising from this study therefore includes: (i) a mesh convergence study focused on the torque-link fillet, (ii) substitution of an aerospace-representative material model with proper yield and ultimate strength allowables, (iii) explicit modeling of the oleo-pneumatic shock strut stiffness/damping behavior, (iv) a transient drop-test simulation per representative landing load factors, and (v) fatigue life estimation at the identified critical fillet under repeated landing cycles.

VI. Conclusion

This paper presented a self-directed static structural finite element study of a twin-wheel nose landing gear assembly, covering CAD assembly modeling, meshing, boundary condition and load definition, and post-processed safety-factor evaluation in Ansys Mechanical. The analysis identified the upper torque-link fillet as the critical location, with a minimum computed safety factor of 0.667 under the assumed static load case, below the conventional unity threshold. While the absolute result must be interpreted cautiously given the simplified material assignment and coarse, unrefined mesh at the critical feature, the study successfully demonstrates a complete and reproducible CAD-to-FEA workflow and identifies concrete, actionable next steps – mesh convergence, material substantiation, and dynamic load simulation – to mature the analysis toward an airworthiness-representative standard. As an independent undergraduate project, this work reflects an initial but structured attempt at applying finite element methods to a genuine aerospace structural problem, and forms a foundation for more advanced graduate-level study in computational structural mechanics and landing gear design.

References

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